VarshaMitra

SYSTEM EXPLAINED
FARMER-CENTERED
IMPROVEMENT PLAN

What the project does, how its guidance is produced, what its limits are, and what to change before adding more features.

Prepared from checked repository documentation, frontend routes/components, and backend service code. Review date: 8 November 2026.

Short answer: VarshaMitra is a Maharashtra-focused weather and crop decision-support system. It combines weather-model data with deterministic climate and crop rules; AI explains approved guidance rather than making the core farming decision.

This report explains and recommends only. No application screens, APIs, or source code have been changed.

1. What this system is

VarshaMitra aims to turn local weather information into a next farming action. A user selects a Maharashtra location, crop, and growth stage. The system combines climate signals and crop rules to offer a prioritized advisory in English or Marathi. Separate officer views support regional monitoring.

Part	What it does
Farmer web app	Dashboard, location/crop selection, forecast and climate views, crop-stage advisory, risk map, Saarthi chat, and supporting audio/soil/mandi/insurance/support cards.
Officer web app	Regional dashboard, risk map and insights for administrative monitoring and drill-down.
Backend	FastAPI resolves locations, retrieves rainfall/weather data, computes indicators and risk, applies crop rules, and returns typed responses.
External information	Open-Meteo weather/reanalysis, OpenStreetMap location/map data, Maharashtra crop/soil references and cited agricultural practices. Coverage and resolution vary.
AI explanation	NuGen/Gemini-backed explanation with deterministic fallback. The validated agronomic action is intended to remain fixed.

Typical farmer journey

1. Choose a village/block or use map/GPS.

2. Select crop and growth stage; the app remembers this context.

3. Backend retrieves forecast or recent model-analysis values for the location.

4. Climate rules compute rain accumulation, dry spell, heavy rain, anomaly/onset and data-quality signals.

5. Crop rules combine signals with crop and stage to prioritize actions.

6. Dashboard/chat explains the reason and caveats; farmer checks actual field conditions before acting.

This is decision support, not a rain gauge, a surveyed farm-boundary map, an official warning service, or a guarantee of crop outcome.

2. How the guidance is produced

The intended recommendation path is systematic and explainable:

Step	System input/processing	What the farmer should understand
Location	Place/PIN/GPS/map click resolves to an administrative block or coordinates. Regional views commonly use block coordinates/centroids.	Nearby grid/block reference, not a sensor at the exact plot.
Weather	Open-Meteo forecast for future weather. The recent-values route can use past-days model analysis, explicitly not rain-gauge observations. ERA5-Land is a separate historical reanalysis.	Label forecast, model analysis, reanalysis, observation, provider and dates distinctly.
Climate signals	Backend summarizes rainfall amounts/windows, dry spells, heavy rain, anomaly, monsoon onset, coverage and freshness.	Show the signal and time window, not only a colored risk label.
Risk	Deterministic rules combine indicators; low coverage can withhold risk or report insufficient data.	Repeatable rules do not by themselves prove forecast accuracy.
Crop advisory	Crop catalogue and growth-stage-specific rules combine weather signals, crop/stage and cited practices to create prioritized actions.	Give the action, why it is relevant now, and when/how to verify locally.
Explanation	NuGen receives canonical advice/evidence; output is validated for required sections and key contradictions. Templates are used if unavailable/rejected.	AI wording may aid comprehension; evidence and field checks stay visible.

Example of a systematic explanation

A code-backed example: the advisory engine treats a single-day rainfall value at or above the 64.5 mm very-heavy-rain threshold shown in its rule as a hazard and can prioritize drainage and postponing field applications. Another rule considers a seven-day dry spell during monsoon. These are **specific triggers**, not proof that every field has the same water condition. Crop, stage, data window, data completeness and local soil moisture still matter.

A trustworthy advisory should show: selected village/block; crop and stage; source and timestamp; whether a value is forecast or model analysis; signal; action/reason; what confidence means; uncertainty; a field check; and a safe local expert-help route.

3. What ?accurate? can and cannot mean

No system can promise perfectly accurate farm-level rainfall or crop guidance. Accuracy should be measured and described in layers, not presented as one reassuring number.

Question	What the repository indicates	Do not imply
How local?	Open-Meteo adapter documents an approximately 11 km grid. Block/centroid is a geographic reference.	Not a sensor reading from the farm; rainfall varies over short distances.
Are recent values observed?	Past-days values from forecast endpoint are explicitly model analysis, not rain-gauge data. ERA5-Land is also modeled reanalysis.	Do not call modeled values local gauge observations.
Is confidence a hit rate?	Risk engine has a data-coverage gate and confidence rationale for completeness/freshness.	High coverage does not prove a forecast is likely correct.
Is soil field-specific?	Regional soil profiles are modeled baselines; a service can also store farmer soil-test data.	Regional soil class is not a lab test of the farmer's plot.
Does AI decide?	Deterministic agronomic rules choose canonical actions; LLM is described as explanation with fallback.	Do not market it as yield prediction or an all-knowing field diagnosis.

How to build systematic trust

- **Label evidence:** forecast, model analysis, reanalysis, sensor/gauge observation, regional estimate, or farmer-entered data.
- **Show freshness:** issue time, valid dates, last updated, missing-data notice, provider and source attribution.
- **Separate confidence:** data completeness, spatial fit, forecast uncertainty and rule applicability are different things.
- **Back-test weather:** compare predictions against quality-controlled gauges/IMD observations across seasons and locations; report error by lead time.
- **Validate farm advice:** agricultural experts review rule versions; evaluate advice with field trials and farmer feedback before claiming outcome improvement.
- **Keep a decision trace:** record inputs/rule version and let farmers correct crop, stage, irrigation and local rainfall.

Safe promise: ?This is the best available local decision support from the displayed data. Forecasts can change; please check your field and local advisories.?

4. Farmer-friendliness: what to change first

Your concern is valid: more charts, cards and indicators do not automatically make a system easier to trust. Keep the decision engine; make the first screen answer a farmer's immediate question.

Challenge	Recommended change
Dashboard feels like a data portal	Make a ?Today on your farm? screen: location/crop; one clear action; short reason; 2-3 steps; ?avoid?; next check time. Put charts behind ?Why this advice??
Technical language	Use everyday Marathi/English first; explain vapsa, anomaly, model and confidence in a short ?What this means? section. No unexplained score.
Incomplete context	Ask essentials first: village, crop, stage/sowing date, rainfed/irrigated. Make land/plot details optional. Confirm selected location visibly.
Advice seems over-certain	Show forecast/estimated/field-check-needed badges, dates, source and uncertainty. Never label modeled rainfall as local gauge data.
Farmers prefer listening	Prominent ?Listen in Marathi?, replay and speed control; optional SMS/voice for low bandwidth.
Connectivity/literacy vary	Large touch targets, icons plus text, high contrast, readable type, low-data pages and cached results visibly marked with age.

Login and biometric sign-in

The checked login is prototype-style: OTP is generated in the browser rather than sent by an SMS provider; farmer records/MPIN are held in a frontend registry, and app state is persisted in browser localStorage. Seeded demo profiles exist. This is not production identity verification. No biometric flow was found in the checked login UI.

- **Do not make biometrics the only entry route.** A farmer may share a phone, change SIM/phone, or use a device without a fingerprint sensor.
- Implement server-verified mobile OTP and recovery first. A PIN needs server-side rate limits, secure storage and device privacy protections.
- Only later consider optional passkeys/WebAuthn with backend verification and explicit consent. VarshaMitra should not collect fingerprints.
- On shared phones, provide visible logout, minimal personal data, private profiles and a shared-device option.

5. Replace the bubble map with an official-style map

The map code creates GeoJSON **Point** features from backend block latitude/longitude and colors points by risk layer. The basemap uses OpenStreetMap raster tiles with terrain/satellite options. A circle around a block coordinate represents that point, not the actual boundary or area of rainfall.

Element	Recommended behavior
Boundaries	Use appropriately licensed authoritative district/taluka/block polygons. Cite publisher, version/date and license; validate projection, topology, names and alignment.
Risk view	Use choropleth fills by a defined block-level risk category; subdued basemap, thin boundary strokes, no oversized bubbles.
Rain view	Separate modeled/observed rainfall from forecast. Show units (mm), period/date and a continuous legend; show probability separately from amount.
Missing data	Use neutral hatch/gray plus ?not enough data?; never color missing data as low risk. Show freshness and provider status.
Interaction	Select block for name, source, time window, rainfall/risk signals, confidence basis and next action. Include search and touch/keyboard-friendly controls.
Farm accuracy	Mark selected village/farm separately from administrative boundaries. Do not invent a parcel boundary.

Suggested map note: ?Block-level rainfall risk for [dates], computed from [source/type]. Administrative boundaries are reference areas; conditions vary within a block.?

Build order: choose and validate polygon source; update GeoJSON points to polygons; define choropleth and neutral no-data style; add title, legend, date, attribution and detail panel; test mobile and accessible colors. If polygons are not available, keep point markers and label them honestly.

6. Prioritized change and feature plan

Priority	Change / feature	Why now	Acceptance check
P0 - now	Simplify farmer home to one daily action, reason, steps, avoid list, source/time and field check. Keep detail views.	Farmers should know what to do before learning a dashboard.	Farmer test: identify action, reason and uncertainty without help.
P0 - now	Make provenance and uncertainty visible; label forecast vs model analysis vs reanalysis.	Trust comes from evidence, not AI branding.	No weather value lacks source, data type and time window.
P1 - foundation	Replace point bubbles with sourced official polygons and professional choropleth.	Directly addresses the map concern.	Polygon source aligns; legend, no-data, period and attribution visible.
P1 - trust	Replace mock login with real, privacy-reviewed sign-in/recovery; retain guest/demo access.	Current frontend OTP/local profile is not secure production authentication.	Server-verified OTP, expiry, rate limit, recovery and shared-device tests.
P1 - inclusion	Make Marathi audio, text size, large controls, low-bandwidth support and stale-data labels prominent.	Fits rural voice and connectivity needs.	Test low literacy/connectivity; stale result cannot look current.
P2 - context	Optional soil health card, irrigation, sowing date and local rainfall correction.	Adds relevant farm context without forcing data entry.	Farmer data is distinguished from regional estimates.
P2 - reach	Opt-in SMS/voice alerts for high-value events; unsubscribe and quiet hours.	Useful when farmers do not open the web dashboard.	Message states village, date, source/type, action and uncertainty.
P2 - feedback	?Was this useful??, field observation and acted/not-acted feedback for agronomy review.	Creates a measured improvement loop.	Consent, minimal data and traceable rule version.

Do not add many cards before simplifying the primary journey. Keep mandi, insurance, emergency support, chat and officer communication as optional tools; verify their source and update timestamp before treating each as a live operational service.

7. Validation checklist and conclusion

Before presenting the system as accurate or ready for everyday farm use, validate technology and farmer experience. These are proposed checks, not existing project results.

- **Usability:** test different districts, rainfed/irrigated farms, ages, phone familiarity and language preference. Ask farmers to set location/crop, understand advice and find source/uncertainty.
- **Action comprehension:** can a farmer explain what to do, why, by when, and what field condition could change it?
- **Forecast verification:** compare against quality-controlled rainfall references by location and lead time; publish limits rather than one accuracy score.
- **Rule review:** record source/institution, rule owner, review date, crop/stage, trigger, exception and version.
- **Map review:** verify polygon provenance/alignment, attribution, legend meaning, no-data handling, mobile layout and accessible colors.
- **Safety/operations:** test offline/stale response, API outage, wrong crop/location, shared-device privacy, account recovery and escalation to KVK/agricultural officers.

What is already valuable

The repository has a strong foundation to preserve: Marathi/English support; crop-stage-aware deterministic advisory rules; explicit data-quality and insufficient-data states; cited weather/agronomy sources; AI explanation constraints and fallback templates; a distinct officer workflow; and accessibility-oriented UI guidance. The recommended work is mainly about making this foundation easier to understand, verify and trust.

Evidence reviewed

README.md; ARCHITECTURE.md; DATA_SOURCES.md; UI_UX_GUIDELINES.md; PHASE7_READINESS.md; farmer dashboard, crop, login and Saarthi routes; FarmerLoginForm and farmer registry; MapLibre risk-map implementation; rainfall/advisory API routes; climate risk and agronomy advisory services; explanation validator/fallback; regional soil service. Repository claims and operational integrations should be independently tested before production claims.

Recommended first three actions: (1) simplify the farmer dashboard, (2) make evidence and uncertainty visible on every advisory, and (3) replace point bubbles with sourced administrative polygons. Do not start with biometric login or more AI features.